

Executive Business Report

Sales Forecasting and Demand Analysis for Superstore Retail

Executive Summary

This project analyzes historical Superstore sales data to improve sales forecasting, inventory planning, and business decision-making. Sales trends, seasonal patterns, anomalies, and product demand were analyzed using data science techniques. Three forecasting models (SARIMA, Prophet, and XGBoost) were compared, and **XGBoost was selected as the best-performing model** because it achieved the lowest prediction errors. An interactive Streamlit dashboard was developed to help business users monitor sales trends, forecasts, anomalies, and product demand without requiring technical knowledge.

1. Key Findings from EDA and Forecasting

- Historical sales show an overall increasing trend with seasonal fluctuations.
 - Sales vary across regions and product categories, indicating different demand patterns.
 - Three forecasting models (SARIMA, Prophet, and XGBoost) were evaluated using **MAE, RMSE, and MAPE**.
 - **XGBoost performed best** with **MAE = 9,397.61**, **RMSE = 12,725.68**, and **MAPE = 15.35%**, making it the final model for deployment.
-

2. Three-Month Sales Forecast

The XGBoost model predicts that sales will continue following the current trend over the next three months. The forecast includes a **95% confidence interval**, which represents the expected range within which future sales are likely to fall. This helps management plan inventory more effectively while reducing the risk of stock shortages or excess inventory.

3. Top Three Anomalies Detected

The anomaly detection model identified unusual sales behaviour that was most likely caused by:

- Promotional campaigns generating unusually high sales.
- Holiday or festival demand spikes.
- Temporary operational or supply chain disruptions.

These observations should be reviewed separately before using them for future forecasting.

4. Product Demand Segmentation

Products were grouped into four demand clusters using K-Means clustering.

Cluster	Recommended Stocking Strategy
Premium Products	Maintain sufficient inventory and monitor demand regularly.
Low Demand Products	Keep limited inventory to reduce holding costs.
High Volume Products	Maintain higher inventory with frequent replenishment.
Fast Growing Products	Increase inventory gradually based on future demand.

This segmentation supports better warehouse utilization and inventory planning.

5. Business Recommendations

1. Use XGBoost for Sales Planning

XGBoost achieved the lowest forecasting errors, making it the most reliable model for inventory planning and purchasing decisions.

2. Prioritize High-Volume Products

Maintain higher stock levels for products in the high-demand cluster to reduce stock-outs and improve customer satisfaction.

3. Monitor Sales Anomalies Regularly

Review unusual sales spikes before updating forecasting models to improve prediction accuracy and operational planning.

6. Risk / Limitation

The forecasting system is based on historical sales data. Unexpected events such as economic changes, supplier delays, or changes in customer behaviour may reduce forecasting accuracy. Regular model retraining with updated data is recommended.

Conclusion

The developed solution combines sales forecasting, anomaly detection, demand segmentation, and an interactive Streamlit dashboard into a practical decision-support system. By selecting **XGBoost as the final forecasting model**, the business can improve inventory management, reduce unnecessary stock, and make better data-driven decisions.